

Roberto Raymundo Gomez Vargas

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Education

Tecnologico de Monterrey (ITESM), Campus Monterrey

B.S. in Robotics and Digital Systems Engineering

Expected Graduation: Dec 2026

Monterrey, Mexico

- Average: 98/100 (Top 1%) | GPA: 4.0/4.0

Coursework: Intelligent Robotics, Computer Vision, Control Systems, Data Structures & Algorithms, Kinematics & Dynamics.

University of Seoul (UOS), Exchange Program — Grade: A+

Aug 2025 – Dec 2025

Coursework: Artificial Intelligence (ML/DL), Advanced Embedded Systems, Solid-State Electronics

Seoul, South Korea

Beijing Institute of Technology (BIT), Summer Program — Grade: A+

Jul 2025 – Aug 2025

Coursework: Advanced Electronics, Embedded Systems, Intelligent Robotics, C++, C Programming

Beijing, China

Work Experience

Robotics Research Intern, AIST – Tsukuba, Japan

Jan 2026 – Sep 2026

- Designed and implemented a modular ROS 2 autonomous-mapping architecture for the CALL-M mobile manipulator, integrating LiDAR/RGB-D fusion, SLAM Toolbox, and Nav2 into a supervised, distributed, frontier-based exploration pipeline.
- Developed two reusable, externally-configurable ROS 2 frameworks, one for autonomous exploration management (target selection, supervision, failure recovery) and one for static–dynamic obstacle filtering (object tracking, velocity estimation).
- Engineered an autonomous ROS 2 door-opening pipeline for a mobile manipulator, integrating RGB-D/LiDAR perception, ML-based door/handle detection, Nav2, and UR5e manipulation; achieved 90% end-to-end success rate across 20 real-robot trials.

Robotics Engineer Intern, Smart Factory MTY – Monterrey, Mexico

Jan 2024 – Jun 2024

- Assembled, diagnosed, and repaired industrial robots (UR3/UR5, xArm 5/6, Baxter, ABB YuMi), improving operational reliability and reducing downtime by 20% via root-cause analysis, corrective repairs, and preventive maintenance.
- Engineered a high-precision end-effector/gripper mounted on an industrial robot (SolidWorks; 3D printing, laser cutting), improving handling repeatability and increasing throughput by 30%, validated through production-cell testing.

Systems & Data Automation Engineer (Part-time), ITESM – Monterrey, Mexico

Feb 2023 – Mar 2026

- Audited compliance documentation and automated databases for university–government contracts across 50 campuses, overseeing \$5.3M in projects; reduced manual work for 100+ employees and improved document processing time by 21%.

Freelance Technology Specialist, Self-Employed – Monterrey, Mexico

Aug 2022 – Present

- Implemented automation and integrated technology for 50+ US/Mexico SME clients, improving operational efficiency.

Research & Invited Talks

[papers: raymundo140.github.io/research]

A ROS 2 Architecture for Autonomous Mapping [Preprint]

Submitted Sep 2026

- First author; paper submitted to IEEE/SICE SII 2027; research conducted at CNRS-AIST JRL, Tsukuba, Japan.

Guest Lecturer, University of Seoul – Seoul, South Korea

Dec 2025

- Invited by the engineering faculty to deliver a lecture to an international cohort at the University of Seoul; presented on neuromorphic computer architectures and led scholarly discussions on the future of hardware-accelerated AI systems.

Projects

[portfolio: raymundo140.github.io]

Self-Driving Car – Manchester Robotics Collaboration

[Project Page](#)

- Developed an autonomous mobile robot for indoor navigation using computer vision, integrating PID line following, real-time traffic-light detection, and YOLOv8 traffic-sign recognition (NVIDIA Jetson Nano, ROS 2, micro-ROS, OpenCV, C++, Python, Git).

Autonomous Tractor Navigation System – John Deere Collaboration

[Project Page](#)

- Led a team to build an autonomous tractor navigation system, estimating position from camera, encoder, and IMU feedback and using PID control to reach wirelessly received target coordinates (STM32, Raspberry Pi, C, Python, FreeRTOS, CAN).

Dual-Elevator Embedded Control System – University of Seoul

[Project Page](#)

- Designed and built a dual-cabin, five-floor elevator system (SolidWorks, 3D printing) with a non-blocking, real-time STM32 controller (C, concurrent FSMs, PWM servo control, Hall-effect safety sensing, and a greedy request-scheduling algorithm).

Skills

- **Programming:** Python, C, C++, C#, MATLAB, Bash, VHDL, Assembly, PyTorch, R, OpenMP, parallel programming.
- **Tools & Hardware:** STM32, Jetsons, Raspberry Pi, FPGAs, LiDAR, RGB-D cams, communication protocols, ROS 2, PCL, tf2, micro-ROS, Nav2, SLAM Toolbox, OpenCV, YOLO, Gazebo, Isaac Sim, Simulink, Linux, Git, Docker, PID, FSMs, SolidWorks.
- **Languages:** Spanish (Native), English (Near-Native – C2), German (Intermediate – B1)